Worksheet 3: Powers, Doubling, and Fast Growth

Topic Focus: Powers of 2, powers of 3, exponential growth, and paper folding.

1. Fold a paper in half once, you get 2 layers. Fold it again, you get 4 layers. How many layers after 5 folds?

A) 10 B) 16 C) 25 D) 32

ans: This is 2 raised to the power of the number of folds. 2 x 2 x 2 x 2 x 2.

2. Look at the sequence: 1, 10, 100, 1000, … What is the rule?

A) Add 9 to previous B) Add 90 to previous C) Multiply previous by 10 D) Square the previous

ans: Each step adds a zero, which means multiplying by the base (10).

3. Bacteria doubles every minute. Start with 1. How many after 4 minutes?

A) 8 B) 16 C) 32 D) 64

Ans: Min 1: 2. Min 2: 4. Min 3: 8. Min 4: 16. (This is 2^4).

4. Powers of 3: 1, 3, 9, 27, 81. Which operation connects each term?

A) Add odd numbers B) Multiply by 3 C) Squaring previous D) Add 6

Ans: Exponential growth means repeatedly multiplying by the same base number.

5. Compare Sequence A (1, 2, 3, 4, 5...) and Sequence B (1, 2, 4, 8, 16...). Which is true?

A) A grows faster B) B grows faster C) Same rate D) B is triangular

Ans: Sequence A is linear (adding). Sequence B is exponential (multiplying). Exponential always outpaces linear.

6. What is the value of 2 raised to the power of 6 (2^6)?

A) 12 B) 32 C) 64 D) 128

ans: Take the 5th power (32) and multiply by 2 one more time.

7. A pond's lily pads double every day. Completely covered on Day 20. When was it HALF covered?

A) Day 10 B) Day 15 C) Day 19 D) Day 18

Ans/hint: If it doubles every day, then working backward, it was half the size exactly one day prior.

8. Which sequence represents the powers of 4?

A) 4, 8, 12, 16... B) 1, 4, 16, 64... C) 1, 4, 9, 16... D) 4, 16, 32, 64...

ans: Multiply by 4 each time. 1x4=4, 4x4=16, 16x4=64.

9. A tree trunk splits into 3 branches. Each of those splits into 3 more. Trunk = Level 1 (1 branch). How many on Level 4?

A) 9 B) 12 C) 27 D) 81

ans: The levels are powers of 3. Level 1: 3^0. Level 2: 3^1. Level 3: 3^2. Level 4: 3^3.

10. What comes before 512 in the sequence of powers of 2?

A) 256 B) 128 C) 1024 D) 250

Ans: To go backward in a doubling sequence, divide by 2.

11. Look at the units digits of the powers of 2 (2, 4, 8, 16, 32, 64...). The last digits form a repeating pattern of:

A) 2, 4, 6, 8 B) 2, 4, 8, 6 C) 2, 4, 8, 2 D) 4, 8, 6, 2

ans: Calculate the first few: 2, 4, 8, 16 (ends in 6), 32 (ends in 2), 64 (ends in 4).

12. A bouncy ball is dropped from 16 meters. Each bounce, it returns to exactly half its previous height. Height after 4 bounces?

A) 1 meter B) 2 meters C) 4 meters D) 8 meters

ans: Divide by 2 four times. 16 -> 8 -> 4 -> 2 -> 1.

13. Which is larger: 2^4 or 4^2?

A) 2^4 B) 4^2 C) They are equal D) Cannot be determined

ans: 2x2x2x2 = 16. 4x4 = 16. This is a rare crossover in exponential numbers.

14. What is the rule for this sequence? 10000, 1000, 100, 10, 1...

A) Divide by 10 B) Subtract 900 C) Divide by 100 D) Multiply by 0.5

Ans: Removing one zero is mathematically equivalent to dividing the number by 10.

15. If a paper is 0.1 mm thick, and you fold it 10 times (doubling thickness each time), roughly how thick is it?

A) 1 mm B) 10 mm C) 100 mm (10 cm) D) 1000 mm (1 meter)

Ans: 2^10 is 1024. 1024 x 0.1 mm = 102.4 mm (approx 10 cm).